

Capstone Project 2: Global Retail “Flash Sale” Resilience

1. Problem Understanding (Architecture Perspective)

This is a **high-scale, real-world distributed system problem** similar to Amazon Prime Day or Flipkart Big Billion Days.

Core Challenges Breakdown

1. Infrastructure Bottlenecks

- Manual scaling → slow response
- Traffic spike → system crash

2. Environment Inconsistency

- Dev ≠ Prod
- Configuration mismatch → deployment failures

3. Security Risks

- Hardcoded credentials
- Untrusted container images

4. Lack of Observability

- No tracing across services
 - Difficult debugging
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Target Architecture (What You Are Building)

A system with:

- GKE (Kubernetes)
 - Terraform (IaC)
 - Private Networking
 - IAM + Workload Identity
 - Auto-scaling
 - Security layers (WAF, Secrets, Image scanning)
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2. SCENARIOS (DETAILED EXPLANATION)

Scenario 1: Traffic Spike (10x in 5 minutes)

Problem

- Sudden load increase
- Pods/VMs insufficient

Solution

- Horizontal Pod Autoscaler (HPA)
 - Cluster Autoscaler
-

Scenario 2: Configuration Drift

Problem

- Manual UI changes
- System breaks

Solution

- Terraform as single source of truth
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Scenario 3: Credential Leak

Problem

- Password exposed in GitHub

Solution

- Secret Manager
 - No hardcoding
-

Scenario 4: Zonal Failure

Problem

- One zone goes down

Solution

- Regional GKE cluster
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3. CASE STUDIES (DEEP + CODE)

Case 1: Infrastructure as Code (Terraform)

Concept Deep Dive

What is IaC?

Infrastructure defined as code:

- Version controlled
 - Repeatable
 - Automated
-

Architecture Components Defined in Terraform

- VPC
 - Subnet
 - GKE Cluster
-

Mini Project Code

provider.tf

```
provider "google" {  
  
  project = "PROJECT_ID"
```

```
    region = "us-central1"

}
```

vpc.tf

```
resource "google_compute_network" "vpc" {

    name          = "retail-vpc"

    auto_create_subnetworks = false

}
```

subnet.tf

```
resource "google_compute_subnetwork" "subnet" {

    name          = "retail-subnet"

    ip_cidr_range = "10.0.0.0/24"

    region        = "us-central1"

    network       = google_compute_network.vpc.id

    secondary_ip_range {

        range_name = "pods-range"

        ip_cidr_range = "10.1.0.0/16"

    }

}
```

```
secondary_ip_range {  
  
  range_name = "services-range"  
  
  ip_cidr_range = "10.2.0.0/20"  
  
}  
  
}
```

gke.tf

```
resource "google_container_cluster" "primary" {  
  
  name = "retail-cluster"  
  
  location = "us-central1"  
  
  
  
  remove_default_node_pool = true  
  
  initial_node_count = 1  
  
  
  
  network = google_compute_network.vpc.name  
  
  subnetwork = google_compute_subnetwork.subnet.name  
  
  
  
  ip_allocation_policy {  
  
    cluster_secondary_range_name = "pods-range"
```

```
services_secondary_range_name = "services-range"

}

}
```

Outcome

- Fully reproducible infra
 - No manual errors
-

Case 2: Multi-Zone High Availability

Concept

Regional Cluster

- Nodes distributed across zones:
 - us-central1-a
 - us-central1-b
 - us-central1-c
-

Terraform Example

```
resource "google_container_cluster" "regional" {

  name    = "retail-regional"
```

```
location = "us-central1" # region, not zone
```

```
node_locations = [
```

```
    "us-central1-a",
```

```
    "us-central1-b",
```

```
    "us-central1-c"
```

```
]
```

```
}
```

Outcome

- Survives zone failure
- High availability

Case 3: Advanced Auto-Scaling

Components

1. HPA (Pod Level)

- Scales based on CPU/memory

2. Cluster Autoscaler

- Adds/removes nodes

Mini Project

Deployment

apiVersion: apps/v1

kind: Deployment

metadata:

name: frontend

spec:

replicas: 3

selector:

matchLabels:

app: frontend

template:

metadata:

labels:

app: frontend

spec:

containers:

- name: app

image: gcr.io/PROJECT_ID/frontend

resources:

requests:

cpu: "200m"

limits:

cpu: "500m"

HPA

kubectrl autoscale deployment frontend \

--cpu-percent=60 \

--min=3 \

--max=50

Outcome

- Handles sudden spikes automatically
-

Case 4: Workload Identity (Zero Trust)

Concept

- No service account keys
 - Secure identity mapping
-

Mini Project

```
gcloud container clusters update retail-cluster \  
  
--workload-pool=PROJECT_ID.svc.id.goog
```

Bind IAM

```
gcloud iam service-accounts add-iam-policy-binding \  
  
sa-name@PROJECT_ID.iam.gserviceaccount.com \  
  
--member="serviceAccount:PROJECT_ID.svc.id.goog[default/app-sa]" \  
  
--role="roles/storage.objectViewer"
```

Outcome

- No credential leaks
 - Secure service access
-

Case 5: Secret Management

Concept

Store sensitive data securely

Mini Project

```
gcloud secrets create db-password --replication-policy="automatic"
```

```
echo "mypassword" | gcloud secrets versions add db-password --data-file=-
```

Kubernetes Integration (Example)

env:

```
- name: DB_PASSWORD
```

valueFrom:

secretKeyRef:

```
name: db-secret
```

```
key: password
```

Outcome

- No secrets in code

Case 6: Edge Security (Cloud Armor)

Concept

- Web Application Firewall (WAF)
-

Example Rule

gcloud compute security-policies create retail-policy

Outcome

- Blocks SQL injection, DDoS
-

Case 7: Image Security

Concept

- Scan container images
-

Solution

- Artifact Registry scanning
- Binary Authorization

Outcome

- Only trusted images deployed

Case 8: Private Networking

Concept

- No public IPs

Terraform Example

```
private_cluster_config {  
  
    enable_private_nodes = true  
  
}
```

Cloud NAT

```
gcloud compute routers create nat-router \  
  
--network=retail-vpc \  
  
--region=us-central1
```

Outcome

- Secure internal communication
-

Case 9: Log-Based Alerting

Concept

- Detect failures early
-

Example

gcloud logging metrics create error_metric \

--description="Error logs" \

--log-filter='severity>=ERROR'

Outcome

- Alerts before users complain
-

Case 10: Drift Detection

Concept

Terraform detects manual changes

Command

terraform plan

Outcome

- Auto-correct infrastructure
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FINAL ARCHITECTURE FLOW

User → Load Balancer → Cloud Armor → GKE → Services → DB

↓

IAM + Secrets + Logging

FINAL BUSINESS IMPACT

Operational Excellence

- 100% automated infrastructure

Scalability

- Handles flash sale spikes

Security

- Zero-trust architecture

Reliability

- Multi-zone resilience